

**UNIVERSITY OF SCIENCE – VNUHCM
FACULTY OF INFORMATION TECHNOLOGY**



Requirements Analysis

Introduction to Software Engineering (CSC13002)

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1 Member Contribution Assessment

ID	Name	Contribution (%)	Signature
24127540	Lê Tấn Thành	100%	
24127234	Lê Song Anh Tài	100%	
23127008	Nguyễn Trọng Tài	100%	
24127479	Nguyễn Bách Nhuận	100%	

2 Problem Statement

Business Problem

Individuals and small businesses increasingly publish content across more than one social media platform to reach their audience, most commonly Facebook and LinkedIn. However, each platform has its own posting interface, login process, and formatting rules, forcing users to repeat the same upload process separately on each one.

For a business with multiple people involved in content creation, this problem is compounded. A content creator may produce a post, but a manager still needs a way to review and approve it before it goes live, and there is currently no single place to do this across both platforms at once. This results in fragmented workflows, duplicated effort, inconsistent publishing schedules, and a lack of centralized oversight over what content has been approved, published, or failed to publish.

The core of the problem lies in several key areas. Content creation and distribution are handled manually and separately per platform, making the process time-consuming and error-prone. There is no standardized review-and-approval workflow for businesses where multiple staff members contribute content, which creates bottlenecks and risks unapproved content going live. Tracking the status of posts (draft, pending review, rejected, published, or failed) across multiple platforms is difficult without a centralized system. Furthermore, there is no unified way to manage connected social accounts, monitor publishing failures (e.g., expired authorization), or analyze audience engagement across channels, limiting users' ability to refine their content strategy.

Target Users

The platform is intended for two main types of end users: Individual users, who manage their own social media presence alone, and Business users, where multiple staff members are involved in producing and publishing content. To support the Business case, the system provides two account roles:

- **Manager (Business):** Acts as the manager of a team. A Manager creates a Workspace, invites Members to join it, connects the business's Facebook and/or LinkedIn accounts, and is responsible for reviewing draft content submitted by Members and deciding whether it can be published.
- **Member (Business):** A subordinate role used by staff within a business. A Member can compose content and submit it for the Manager's review but cannot publish directly.
- **Individual:** A user working alone who has the same feature access as a Manager but without any team-related features, since there is no team to manage. An Individual reviews and publishes their own content directly, without needing approval from anyone else.

Operating Environment

The proposed software solution is designed to function within a specific operating environment.

- **Client-Side:** The application will require a modern web browser that supports HTML5, CSS3, and JavaScript (e.g., Google Chrome, Firefox, Edge), with no additional client-side installation required.
- **Server-Side:** The back-end will run as a server-side application exposing RESTful endpoints, served through an ASGI server (e.g., Uvicorn), enabling asynchronous handling of I/O-bound operations such as external API calls and scheduled publishing.
- **Database:** The system will connect to a PostgreSQL relational database to store users, workspaces, connected accounts, posts, and review/publishing statuses.
- **External Services:** The system integrates with the Facebook Graph API and the LinkedIn API for OAuth 2.0 authentication and content publishing, and with a self-hosted n8n automation instance to handle asynchronous multi-platform distribution.

Design and Implementation Constraints

The development of this solution will be governed by several key constraints.

- **Programming Language:** The front-end will utilize JavaScript with React and Vite, styled with Tailwind CSS, while the back-end will be developed using Python with FastAPI (or Django REST Framework).
- **Architecture:** The system shall be implemented as a client-server web application with a clearly separated front-end and back-end, ensuring all business rules (such as the approval workflow) are enforced solely on the back-end.
- **Security:** The system will use hashed passwords and JWT-based session tokens transmitted only over HTTPS. Access tokens obtained via OAuth 2.0 must be stored encrypted at rest and never displayed in plain text. Role-based access control (RBAC) must be enforced at the back-end level so that only Managers (or Individuals acting on their own content) can approve and publish posts.
- **Scalability:** The system must be designed using a decoupled, modular architecture to accommodate future growth in users, workspaces, and connected platforms, though production-grade scalability is out of scope for this course project.
- **Data Integrity:** The relational database must enforce integrity between workspaces, members, posts, and their review statuses through foreign keys and transactions.
- **Documentation:** The project will produce standard software development lifecycle documentation, including API documentation via Swagger/OpenAPI and a system user manual.
- **Deployment:** As this is a school-based project, no production deployment infrastructure is guaranteed; the full stack (front-end, back-end, database, and automation node) shall be containerized and orchestrated locally via a unified docker-compose.yml.

3 Requirements Overview

3.1 Stakeholders

STT	Stakeholder	Description
1	Project Development Team	The Project Development Team is responsible for transforming the approved requirements and high-fidelity prototypes into the Omni Platforms web application. Its responsibilities include implementing the separated front-end and back-end architecture, enforcing role-based approval and publishing rules at the back-end level, integrating PostgreSQL, n8n, external AI services, Facebook, and LinkedIn APIs, and producing API and user documentation. The team requires stable, validated requirements and technically feasible user flows before implementation. Its influence and interest are both high because its architectural, security, performance, and integration decisions directly determine whether the requirements of all other stakeholders can be satisfied.
2	Manager - Business User	Managers administer the content-production process within a Business Workspace. They create and configure the workspace, process Member access, assign tasks, manage connected Facebook and LinkedIn channels, monitor team KPIs, and review content submitted by Members. Managers can comment on, edit, approve, or reject submitted drafts and are responsible for authorizing content before it is distributed to external platforms. Their main needs are centralized oversight, secure publishing control, clear approval queues, reliable notifications, audit logs, channel-permission management, and team-performance information. Their influence and interest are high because they control workspace membership, distribution permissions, and the final approval decision for business content.
3	Member - Business Content Creator	Members are staff members who participate in a shared Business Workspace by entering a valid Workspace ID. They create and edit draft posts, manage prompt templates and knowledge-base contexts, use

		AI and GEO/SEO assistance, select authorized distribution channels, and submit completed content for Manager review. Members can receive rejection feedback, revise their drafts, and track their assigned tasks and submission statuses, but they cannot configure social-media connections or publish directly. Their main needs are efficient content-authoring tools, clear task information, access to approved channels, timely review feedback, and transparent post-status tracking. Their interest is high, while their influence is medium because they create the content but do not hold final publishing authority.
4	Individual User	Individual Users are independent content creators who manage their own social-media presence without participating in a team workspace. They can connect their own Facebook or LinkedIn accounts, manage prompts and contexts, generate or manually write content, apply GEO/SEO optimization, select target channels, publish posts directly, and review cross-platform analytics. Unlike Members, they do not pass through the Pending Review or Rejected stages and have full control over their own content lifecycle. Their main needs are an autonomous and efficient multi-channel workflow, reduced repetitive posting work, reliable account connections, direct publishing, clear failure reporting, and consolidated engagement analytics. Their interest and operational influence are high because they independently make all decisions relating to their own content.
5	Social Media Platform Providers - Meta/Facebook and LinkedIn	Meta/Facebook and LinkedIn provide the external OAuth 2.0 authentication, publishing, and engagement-metrics APIs used by Omni Platforms. The system depends on these providers to authorize connected accounts, accept publishing payloads, return live post identifiers and URLs, and provide interaction data such as likes, comments, shares, and impressions. The project must comply with their API policies, securely manage access tokens, use the required data formats, and handle token expiration, network errors, and API rate limits. Their influence is high because their authorization rules, API availability, and responses determine whether posts and analytics can be processed successfully. Their direct interest in this academic project is comparatively low.

6	External AI Content Generation Service Provider	The External AI Content Generation Service Provider supplies the generative capability used by Omni Platforms to produce draft text from selected prompt templates and contextual knowledge-base data. The back-end sends structured prompt and context information to the service and returns the generated response to the rich-text editor, where users can review and modify it. The integration requires secure request handling, reliable responses, and contextually relevant generated content. Its influence is medium because service quality and availability affect AI-assisted drafting; however, the platform still allows users to disable AI assistance and write content manually. Its direct interest is lower than that of the primary users.
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3.2 Requirements

3.2.1 Functional Requirements Specification

3.2.1.1 Authorization Module

FR-AUTH-01: Sign Up

- The system shall allow users to register a new account using an email address and password.
- The system shall send a verification email when a user registers with an email account.
- The system shall support three user roles:
 - **Individual User**
 - **Manager**
 - **Member**
- Members shall be allowed to create and manage prompts, contexts, and draft content but shall not publish content directly.
- Managers shall have exclusive privileges to review, comment on, approve, and reject draft content submitted by Members.
- Individual Users shall have full control over their own content lifecycle and may publish content without requiring Manager approval.

FR-AUTH-02: Log In / Log Out

- The system shall allow registered users to log in using their credentials.
- The system shall support the **Remember Me** feature.
- The system shall automatically terminate inactive sessions after a predefined timeout period.

FR-AUTH-03: Forgot/Reset Password

- The system shall allow users to reset their password through a verification link sent to their registered email address.

FR-AUTH-04: User Profile Management

The system shall allow users to update their personal information, including:

- Display Name
- Avatar
- Time Zone

FR-AUTHZ-05: Authorization & Access Control

- The system shall securely validate JWT session tokens over HTTPS.
- The system shall ensure that users can access only the features and resources permitted by their assigned role.
- The system shall restrict unauthorized access to protected frontend routes and backend services.

3.2.1.2 Workspace Module**Manager Functions****FR-WORK-MGR-01: Create Workspace**

- The system shall allow a Manager to create a new business workspace.
- The system shall generate a unique Workspace ID for every workspace.
- The system shall assign the creator as the workspace administrator.
- The system shall store the workspace information and confirm successful creation.

FR-WORK-MGR-02: Workspace Management

- The system shall allow Managers to update workspace information and workspace settings.
- The system shall validate all required information before saving changes.

FR-WORK-MGR-03: Team Management

- The system shall allow Managers to manage Members within their workspace.
- The system shall allow Managers to approve or remove Members when necessary.

FR-WORK-MGR-04: Task Assignment

- The system shall allow Managers to assign tasks and deadlines to Members within the workspace.
- Assigned tasks shall appear in the Member's task list and calendar.

FR-WORK-MGR-05: Team Calendar

- The system shall provide a shared calendar displaying assigned tasks and upcoming deadlines.
- The calendar shall allow Managers to review the team's schedule.

FR-WORK-MGR-06: Team Performance Monitoring

- The system shall provide Managers with an overview of team performance and task progress.

FR-WORK-MGR-07: Content Review & Approval

- The system shall allow Managers to review draft content submitted by Members.
- Managers shall be able to comment on, approve, or reject submitted drafts.
- Only approved content shall proceed to the publishing process.

Member Functions**FR-WORK-MEM-01: Join Workspace**

- The system shall allow Members to join a workspace by providing a valid Workspace ID and Workspace Password.
- The system shall verify the provided workspace credentials before granting access.
- The system shall create the membership association after successful verification.

FR-WORK-MEM-02: Prompt & Context Management

- The system shall allow Members to create, update, and use prompts and contexts for content creation.

FR-WORK-MEM-03: Draft Management

- The system shall allow Members to create, edit, and save draft posts within the workspace.

FR-WORK-MEM-04: Publication Restriction

- Members shall not be allowed to publish content directly.
- The system shall require all Member-created posts to be submitted for Manager review before publication.

3.2.1.3 AI Post Module**FR-POST-01: AI-Assisted Content Generation**

- The system shall allow users to select prompts and contexts for AI-assisted content generation.
- The system shall generate draft content using the configured AI service.
- The generated content shall be displayed in the rich-text editor.
- AI-generated posts shall automatically be labeled "**By AI**".

FR-POST-02: Rich-Text Editing

- The system shall provide a rich-text editor supporting:
 - Text formatting
 - Images
 - Hyperlinks

- The system shall allow users to modify generated content before publication.
- The system shall allow users to configure GEO and SEO information.
- The system shall require users to select at least one target distribution channel before publication.

FR-POST-03: Post Lifecycle Management

The system shall support the following post statuses:

Status	Description
Draft	Initial state for newly created posts.
Pending Review	Waiting for Manager approval.
Rejected	Returned to the author for revision.
Ready for Distribution	Approved and ready for publishing.
Published	Successfully published.
Failed	Publishing unsuccessful.

FR-POST-LC-01: Business Workflow

- Posts created by Members shall follow the workflow:

Draft → Pending Review → Ready for Distribution → Published

or

Draft → Pending Review → Rejected → Draft

FR-POST-LC-02: Individual Workflow

- Posts created by Individual Users shall bypass the review process and follow:

Draft → Ready for Distribution → Published

3.2.1.4 Distribution Module**FR-DIST-01: Social Platform Connection**

- The system shall allow users to connect supported social media accounts through OAuth authorization.
- The system shall allow users to add, remove, and reconnect social media accounts.

FR-DIST-02: Content Publishing

- Only posts with the **Ready for Distribution** status shall be eligible for publishing.

- The system shall publish approved posts to the selected social media platforms.
- The system shall support publishing to Facebook and LinkedIn.

FR-DIST-03: Publishing Result Processing

- The system shall update the publication status after receiving the publishing result.
- Successfully published posts shall be marked as **Published**.
- Failed publishing attempts shall be marked as **Failed**.
- The system shall record publishing errors and notify users when publishing fails.

3.2.2 Non-Functional Requirements Specification

3.2.2.1 Security Requirements

Requirement ID	Category	Detailed Description
SEC-01	Authentication	The system shall require users to authenticate using a valid account before accessing protected resources or performing authenticated operations. Authentication may be implemented using JWT-based sessions or OAuth 2.0 where applicable.
SEC-02	Authorization	The system shall enforce Role-Based Access Control (RBAC). In a Business Workspace, only Managers shall be permitted to approve and publish content, while Members shall only be permitted to create and submit drafts for review. Individual Users shall be permitted to publish their own content directly.
SEC-03	Data in Transit	All communication between clients and the server shall be encrypted using HTTPS (TLS 1.2 or higher).
SEC-04	Password Storage	User passwords shall never be stored in plaintext. The system shall store passwords using a secure one-way hashing algorithm (e.g., Argon2 or bcrypt) with a unique salt.
SEC-05	Input Validation	The system shall validate and sanitize all user inputs on the server side to reduce security risks

such as SQL Injection and Cross-Site Scripting (XSS).

3.2.2.2 Performance Requirements

Requirement ID	Category	Detailed Description
PER-01	Response Time	The system shall display the main dashboard within 2 seconds under normal operating conditions. Standard API requests shall return responses within 500 milliseconds on average.
PER-02	AI Generation Response	After a user submits a prompt for AI-assisted content generation, the system shall return the generated content to the editor as soon as the external AI service responds.
PER-03	Task & Notification Delivery	Workspace notifications, task assignments, and review status updates shall be delivered to connected users within 1 second after the corresponding action is completed.
PER-04	File Upload	The system shall support uploading images, video up to 50 MB per file for post creation.
PER-05	Concurrent Users	The system should support at least 500 concurrent users without significant degradation of response time.
PER-06	Database Optimization	Frequently accessed data, such as dashboards, calendars, and post lists, should be optimized through indexing and efficient database queries.

3.2.2.3 Usability

Requirement ID	Category	Detailed Description
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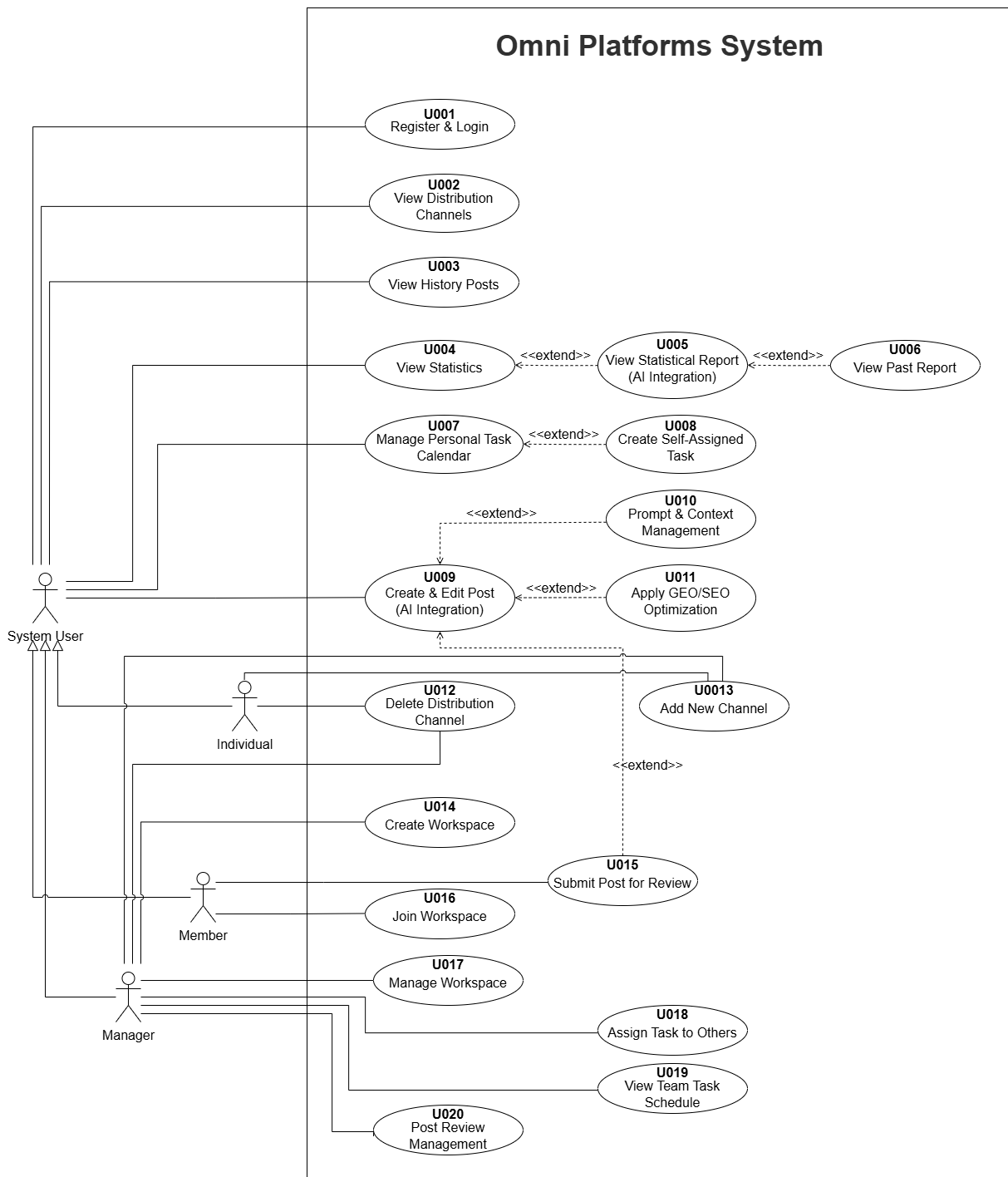
USA-01	Navigation	The system shall provide a consistent navigation structure across all modules.
USA-02	Calendar	The calendar shall clearly display scheduled tasks, deadlines, and assigned members.
USA-03	Editor	The rich-text editor shall support inserting text, images, and hyperlinks.

3.2.2.4 Reliability

Requirement ID	Category	Detailed Description
REL-01	Data Integrity	The system shall preserve post and workspace data during normal operations.
REL-02	Failure Recovery	If content publishing fails, the system shall record the failure and retain the post for future retry.

4 Requirements Analysis

4.1 Use Case model



4.2 Use Case Specification

4.2.1. Use Case 1

<i>Use case ID</i>	U001
<i>Use Case</i>	Register & Login
<i>Brief Description</i>	Describe the process for a new user to register a system account (choosing either the Individual or Business branch) and sign in to the application to obtain a JWT session token.
<i>Actor</i>	System User
<i>Precondition</i>	Users access the Omni Platforms website via a browser that supports HTML5 using the secure HTTPS protocol.
<i>Result</i>	Once the user is successfully authenticated, the system generates a JWT code, saves it to the client-side, and redirects them to the correct dashboard interface according to their role.
<i>Main Scenario</i>	1. The user selects the "Sign up" feature on the desktop. 2. The system displays an information entry form including: Email, Password, Username, and Account Type selection (Individual or Business). 3. The user enters all valid information and chooses suitable roles then clicks "Sign up". 4. The system performs password hashing, saves the account information to the PostgreSQL database, and displays a successful registration notification. 5. The system redirects the user to the Login page. 6. The user enters the email and password they've just created and clicks "Login". 7. The system compares login information with the database. 8. The system generates Session Tokens in JSON Web Token (JWT) format containing role information (Role Payload) and sends them back to the Client. 9. The browser stores the security token and redirects the user to the corresponding internal dashboard page.
<i>Alternative Scenarios</i>	[At step 3] The user enters a duplicate email address that already exists in the system: 3a. System for detecting duplicate primary keys in the database.

	3b. The system prevents registration, displays an error message "Email already registered," and requests re-entry. [At step 7] The user entered the wrong Email or Password when logging in: 7a. The authentication system failed. 7b. The system displays a "Invalid credentials" warning, does not issue JWT tokens, and asks the user to double-check their information.
<i>Non-Functional Constraints</i>	Confidentiality: Passwords stored in the database must be securely hashed. Security: The JWT token encryption code transmitted and received over the network must run entirely on HTTPS

4.2.2. Use Case 2

<i>Use case ID</i>	U002
<i>Use Case</i>	View Distribution Channels
<i>Brief Description</i>	Users view the list of authorized social accounts/channels and their current connection status
<i>Actor</i>	System User
<i>Precondition</i>	User is logged in and redirected to their corresponding dashboard
<i>Result</i>	Full display of active/inactive authorized distributing channels (Facebook/LinkedIn)
<i>Main Scenario</i>	1. User navigates to the "Distribution" module located on the left sidebar of the web 2. System retrieves linked channels from the database 3. User Interface displays all names, platform icons, and status badges of distributable channels
<i>Alternative Scenarios</i>	[At step 3] No preconfigured channels/accounts" 3a. For Member role: System displays a "No channels authorized. Please consult your Manager." prompt For the others: System displays a "No channels connected" prompt and "Add" button
<i>Non-Functional Constraints</i>	Performance: Fetching the list must respond in under 500ms

4.2.3. Use Case 3

<i>Use case ID</i>	U003
<i>Use Case</i>	View History Posts
<i>Brief Description</i>	Users browse a chronological archive of all previous post activities
<i>Actor</i>	System User
<i>Precondition</i>	User is logged in with an active account
<i>Result</i>	List of historical posts with status filters is displayed
<i>Main Scenario</i>	1. Users access the "Post Management" section. 2. The system fetches post logs from PostgreSQL 3. User interface displays posts title, platform icons, and status (Published, Failed, Pending, Approved, Rejected, etc. - the number of filters correspondingly based on their roles)
<i>Alternative Scenarios</i>	[At step 3] No recorded logs 3a. User interface displays "No activity found" screen
<i>Non-Functional Constraints</i>	Performance: Queries must be indexed to maintain performance and optimize response times

4.2.4. Use Case 4

<i>Use case ID</i>	U004
<i>Use Case</i>	View Statistics
<i>Brief Description</i>	This feature describes how to scrape post performance data and aggregate it into visual statistical charts, helping users assess reader engagement across different social media platforms
<i>Actor</i>	System User
<i>Precondition</i>	The system already has articles that have been successfully published on social media.
<i>Result</i>	The system outputs a visual chart and several statistical outputs showing the total interaction volume in general from all channels to the screen.
<i>Main Scenario</i>	1. Users access the "Statistics" section. 2. The system calls the Interaction Analysis module to scan the list of posts with the status "Published" in the database.

	3. This module sends requests to retrieve telemetry data from real articles via the online post ID, connecting to the Facebook Graph API and LinkedIn Metrics API. 4. The system receives raw data of interaction in general 5. The system aggregates and calculates metrics and creates visual graphs (bar chart comparing Facebook vs LinkedIn interactions, weekly growth trend line chart) which are displayed on the ReactJS UI via graphics libraries.
<i>Alternative Scenarios</i>	[At step 3] The third-party social network API temporarily blocks interaction data scraping or is too slow to respond: 3a. The system is unable to retrieve the latest real-time data. 3b. The system will retrieve old interactive data stored in the cache - if any - database from the previous scan and display it temporarily with the message: "Displaying cached data. Live metrics temporarily unavailable".
<i>Non-Functional Constraints</i>	Usability: The statistical chart interface must be designed in a responsive format, displaying accurately and smoothly on both desktop PC and laptop screens with a minimum resolution of 1024x768. Performance: Fetching the data and graphical outputs must complete in under 3 seconds

4.2.5. Use Case 5

<i>Use case ID</i>	U005
<i>Use Case</i>	View Statistical Report (AI Integration)
<i>Brief Description</i>	Extends U004 to pull real-time telemetry from third-party networks and generate AI-driven performance insights
<i>Actor</i>	System User
<i>Precondition</i>	User is currently inside the active U004 Statistics view.
<i>Result</i>	An advanced analytical report integrated with AI engagement predictions is outputted.
<i>Main Scenario</i>	1. Inside the U004 view, the user selects the "AI Assistant Support" or "Deep Analysis Report" module. 2. System sends real-time scraping requests to Facebook Graph API and LinkedIn Metrics API using online post IDs. 3. The system receives the raw interaction metrics.

	4.Data payloads are sent to the AI Service to package deep analysis summaries. 5.Advanced reports containing text recommendations alongside bar charts and trend lines are rendered on the ReactJS UI.
<i>Alternative Scenarios</i>	[At step 2] If third-party APIs block scraping or time out: System retrieves cached data from the previous scan and displays a "Displaying cached data. Live metrics temporarily unavailable" warning. [At step 5] User requests comparison with legacy summaries: System extends to U006 (View Past Report).
<i>Non-Functional Constraints</i>	Performance: Processing complex AI analytical text queries must execute via background worker queues to avoid UI blocking.

4.2.6. Use Case 6

<i>Use case ID</i>	U006
<i>Use Case</i>	View Past Report
<i>Brief Description</i>	Extends U005 to allow users to review and compare previously saved AI statistical reports.
<i>Actor</i>	System User
<i>Precondition</i>	The user is inside the U005 AI Statistical Report dashboard.
<i>Result</i>	A historical report snapshot is retrieved and loaded onto the interface.
<i>Main Scenario</i>	1. Inside U005, the user clicks the "View Past Reports Archive" button. 2. System queries the database for archived analytical snapshots matching the workspace profile. 3. User selects a specific timestamp row from the generated log list. 4. System retrieves the archived AI conclusions and displays them for historical comparison.
<i>Alternative Scenarios</i>	[At step 2] No historical AI reports are found in the database for the given profile context: 2a. The system stops populating the report history list block. 2b. UI displays an informational prompt: "No archived AI reports found for this workspace."

<i>Non-Functional Constraints</i>	Usability: Historical report tables must provide sorting filters based on dates
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4.2.7. Use Case 7

<i>Use case ID</i>	U007
<i>Use Case</i>	Manage Personal Task Calendar
<i>Brief Description</i>	System users view and organize their publication workflows using an interactive personal calendar.
<i>Actor</i>	System User
<i>Precondition</i>	The user is logged in with an active account status.
<i>Result</i>	Personal workspace timeline displays scheduled post workflows.
<i>Main Scenario</i>	1. The user navigates to the "Calendar" module on the main navigation panel. 2. System queries active post tasks assigned to or created by the user. 3. UI displays a calendar grid layout populated with task indicator blocks.
<i>Alternative Scenarios</i>	[At step 2] User decides to plot a new item on the timeline: System extends to U008 (Create Self-Assigned Task).
<i>Non-Functional Constraints</i>	Performance: The personal calendar state should fetch and sync updates within 800ms of loading.

4.2.8. Use Case 8

<i>Use case ID</i>	U008
<i>Use Case</i>	Create Self-Assigned Task
<i>Brief Description</i>	Extends U007, allowing system users to instantiate a private task entry directly from the calendar UI.
<i>Actor</i>	System User
<i>Precondition</i>	The user is interacting with the personal task calendar dashboard (U007).
<i>Result</i>	A new personal task is created and displayed on the calendar.

<i>Main Scenario</i>	1. Inside U007, the user clicks on a calendar date cell or clicks "Add New Task". 2. The system displays a modal dialog box asking for Title, Description, and Deadline. 3. The user fills out required fields and clicks "Confirm". 4. The system saves the entry into PostgreSQL and dynamically updates the calendar grid.
<i>Alternative Scenarios</i>	[At step 3] The user leaves the mandatory "Task Title" input blank: 3a. System blocks submission and flags the input box border in red. 3b. UI displays an inline validation note: "Task Title is a mandatory field." [At step 3] The user picks a target deadline date/time that is in the past: 3a. System flags the date picker component. 3b. UI displays an error message: "Cannot schedule tasks in past timelines."
<i>Non-Functional Constraints</i>	Usability: The task creation module must have input validation helpers preventing empty task headers.

4.2.9. Use Case 9

<i>Use case ID</i>	U009
<i>Use Case</i>	Create & Edit Post (AI Integration)
<i>Brief Description</i>	Writing and editing workflow for social media posts, with options to call helper modules.
<i>Actor</i>	System User
<i>Precondition</i>	The user has successfully logged into the system.
<i>Result</i>	A complete post entity is successfully created in Draft format.
<i>Main Scenario</i>	1. The user navigates to the "Content" module 2. The system displays a multimedia content editor interface (Rich-text Post Editor) along with a target platform list (Facebook, LinkedIn). 3. The user manually writes text captions into the canvas area. 4. The user attaches illustrative media assets from their local machine and checks desired channels.

	5. The user clicks the "Save Draft" button to store the record in PostgreSQL.
<i>Alternative Scenarios</i>	[At step 3] User selects "AI Assistant Support": System extends to U010 (Prompt & Context Management). [At step 3] User wants to verify SEO structures: System extends to U011 (Apply GEO/SEO Optimization). [At step 5] A Member wants to submit the post for manager approval: System extends to U015 (Submit Post for Review).
<i>Non-Functional Constraints</i>	Usability: The rich-text editor must support smooth display of special characters, line breaks, and popular social media emojis.

4.2.10. Use Case 10

<i>Use case ID</i>	U010
<i>Use Case</i>	Prompt & Context Management
<i>Brief Description</i>	Extends U009 to package preset prompt styles and statistics context for AI text content generation.
<i>Actor</i>	System User
<i>Precondition</i>	The user is actively working inside the U009 Create & Edit Post interface.
<i>Result</i>	Generated AI text is loaded directly into the active editing text box.
<i>Main Scenario</i>	1. Inside U009, the user clicks the "AI Assistant Support" button. 2. System requests the user to select a command prompt template and toggle data contexts. 3. User defines parameters; system packages information and dispatches it to the AI Content Generation Service. 4. System receives the structured response text asynchronously. 5. System inserts the AI generation directly into the active U009 editing frame.
<i>Alternative Scenarios</i>	[At step 4] The AI Generation remote service endpoint times out or returns an empty payload: 4a. Backend catches the response code mismatch. 4b. System pops an informational notification alert box: "AI Generation failed to respond. Your manual text remains untouched."

<i>Non-Functional Constraints</i>	Performance: The AI text block must render streaming or fully complete text in the editor window in under 4 seconds.
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4.2.11. Use Case 11

<i>Use case ID</i>	U011
<i>Use Case</i>	Apply GEO/SEO Optimization
<i>Brief Description</i>	Extends U009 to optimize keywords and metadata structure for next-generation AI Search visibility.
<i>Actor</i>	System User
<i>Precondition</i>	The user is actively writing or editing inside the U009 interface.
<i>Result</i>	Optimal structures and hashtag bundles are appended to the post canvas
<i>Main Scenario</i>	1. During the writing process in U009, the user clicks the "Optimize SEO/GEO" button. 2. The system extracts text content from the active editor buffer. 3. The system runs matching checks against ranking metric databases. 4. UI presents a recommendations drawer showing optimization scoring metrics and trending hashtags. 5. User clicks "Apply Improvements", prompting the system to insert optimized hashtag arrays into the text area. 6. User returns to the main editing interface of U009.
<i>Alternative Scenarios</i>	[At step 3] If text is under 20 characters: System halts analysis and displays "Content too short for SEO analysis. Please write more content first."
<i>Non-Functional Constraints</i>	Performance: The analysis filters must run asynchronously in the background without causing lag or freezing the ReactJS UI while typing.

4.2.12. Use Case 12

<i>Use case ID</i>	U012
<i>Use Case</i>	Delete Distribution Channel
<i>Brief Description</i>	Disconnects and removes authorized social media channel configurations from the system profile.

<i>Actor</i>	Manager, Individual
<i>Precondition</i>	The user is navigating inside the U002 Distribution interface panel.
<i>Result</i>	Target OAuth connection credentials are expunged from database layers.
<i>Main Scenario</i>	1. The user enters the account distribution settings area. 2. The user clicks the "Delete/Disconnect Channel" button mapped to a specific platform card. 3. The system displays a warning prompt asking to confirm the disconnection. 4. The user confirms the choice. 5. FastAPI updates records, deleting credential rows out of the PostgreSQL database.
<i>Alternative Scenarios</i>	[At step 4] The selected distribution channel has posts currently marked in the "Pending Review" or "Ready for Distribution" state: 4a. System intercepts database operation routine checks. 4b. UI displays an error dialogue: "Cannot disconnect channel. Active posts are currently queued for this platform. Please clear or re-route posts first."
<i>Non-Functional Constraints</i>	Security: Deleting a channel record must immediately wipe all cached access keys and access tokens from servers.

4.2.13. Use Case 13

<i>Use case ID</i>	U013
<i>Use Case</i>	Add New Channel
<i>Brief Description</i>	Connects external network social profiles into the system via official authorization pathways.
<i>Actor</i>	Manager, Individual
<i>Precondition</i>	The user is navigating inside the Distribution panel interface.
<i>Result</i>	External OAuth tokens are captured and written to database tables.
<i>Main Scenario</i>	1. The user clicks the "Add New Channel" button. 2. System initializes external link logic redirections pointing to external login workflows (Facebook Page / LinkedIn Profile OAuth).

	3. The user successfully signs into their social profile and accepts access request scopes. 4. The system captures security callback data, writes access keys into PostgreSQL, and refreshes dashboard components.
<i>Alternative Scenarios</i>	[At step 3] User rejects app permission scopes or clicks "Cancel" on external OAuth window screens: 3a. Platform authorization redirects callback with access denied indicators. 3b. System catches failure states and displays message: "Integration cancelled by user. Channel permissions were not granted."
<i>Non-Functional Constraints</i>	Security: All authentication callbacks must handle state query parameters to protect against CSRF attacks.

4.2.14. Use Case 14

<i>Use case ID</i>	U014
<i>Use Case</i>	Create Workspace
<i>Brief Description</i>	Creates a separate workspace for managing projects and team members.
<i>Actor</i>	Manager
<i>Precondition</i>	The Manager has successfully registered an account and is logged in but has not yet created a workspace.
<i>Result</i>	A new workspace is created, the Manager is assigned as its administrator, and the workspace is ready for use.
<i>Main Scenario</i>	1. The Manager signs in after successfully registering a Business account. 2. The system prompts the Manager to create a workspace before accessing other features. 3. The Manager enters the required workspace information and submits the form. 4. The system validates the submitted information. 5. The system generates a unique Workspace ID, creates the workspace, and assigns the Manager as its administrator. 6. The system redirects the Manager to the workspace dashboard.

<i>Alternative Scenarios</i>	[At Step 3] The Manager submits the form without entering all required information. 3a. The system validates the submitted fields. 3b. The system highlights the missing required fields and displays the message: "Please complete all required fields before continuing."
<i>Non-Functional Constraints</i>	Confidentiality: Strict database access isolation must ensure that the metadata of the newly created workspace cannot be accessed by unauthorized tenants.

4.2.15. Use Case 15

<i>Use case ID</i>	U015
<i>Use Case</i>	Submit Post for Review
<i>Brief Description</i>	Extends U009 to push drafts into a manager's moderation queue, locking member-level editing access.
<i>Actor</i>	Member
<i>Precondition</i>	Post is currently in the default "Draft" status inside the U009 workflow.
<i>Result</i>	Content is locked at the client side, and the post status maps to "Pending Review".
<i>Main Scenario</i>	1. Inside U009, the Member clicks the "Submit for Review" button. 2. System checks validity parameters (Text string presence, target distribution channel selection). 3. The system locks down the editor view framework on the client interface. 4. The system shifts the status field row to "Pending Review" inside the database. 5. Notification APIs push a real-time banner alert badge onto the dashboard bell icon of the Manager.
<i>Alternative Scenarios</i>	[At step 2] If target channels are missing: System halts submittal, displaying a red warning: "Please select at least one target social platform before submitting."
<i>Non-Functional Constraints</i>	Security: Editing lockout rules must enforce validation checks at both the UI presentation layer and the Backend API middleware layer to block manual API intercept injections (e.g., via Postman).

4.2.16. Use Case 16

<i>Use case ID</i>	U016
<i>Use Case</i>	Join Workspace
<i>Brief Description</i>	Allows a Member to join an existing workspace by providing the workspace credentials assigned by the Manager.
<i>Actor</i>	Member
<i>Precondition</i>	The Member has received a valid Workspace ID and Workspace Password from the Manager
<i>Result</i>	The Member successfully joins the target workspace.
<i>Main Scenario</i>	1. The Member selects the Business account type and the Member role. 2. The Member enters their username, email address, Workspace ID, and Workspace Password. 3. The Member submits the registration form. 4. The system validates the Workspace ID and Workspace Password. 5. The system associates the Member account with the specified workspace. 6. The system completes the registration and grants the Member access to the workspace.
<i>Alternative Scenarios</i>	[At Step 4] The Workspace ID or Workspace Password is invalid. 4a. The system rejects the registration request. 4b. The system displays the message: Invalid Workspace ID or Workspace Password. [At Step 5] The workspace has reached its member capacity. 5a. The system cancels the join request. 5b. The system displays: "This workspace cannot accept additional members."
<i>Non-Functional Constraints</i>	Security: The system shall validate the Workspace ID and Workspace Password before allowing a Member to join a workspace.

4.2.17. Use Case 17

<i>Use case ID</i>	U017
<i>Use Case</i>	Manage Workspace

<i>Brief Description</i>	Allows the Manager to manage workspace information, team members, and workspace settings.
<i>Actor</i>	Manager
<i>Precondition</i>	The Manager is authenticated and is the administrator of the workspace.
<i>Result</i>	The workspace settings are successfully updated.
<i>Main Scenario</i>	1. Manager opens Team Workspace. 2. System displays the current workspace information and member list. 3. Manager updates workspace information or manages team members. 4. Manager saves the changes. 5. System validates and applies the updates.
<i>Alternative Scenarios</i>	[At Step 3] The Manager submits the configuration without completing all required fields. 3a. The system validates the configuration fields. 3b. The system highlights the missing or invalid fields and displays the message: "Please correct the highlighted fields before saving changes "
<i>Non-Functional Constraints</i>	Security: The entire workspace management interface must be strictly protected through permissioned Role-Based Access Control (RBAC) middleware layers.

4.2.18. Use Case 18

<i>Use case ID</i>	U018
<i>Use Case</i>	Assign Task to Others
<i>Brief Description</i>	Managers design and assign specific content publishing tasks or deadlines to associated team members.
<i>Actor</i>	Manager
<i>Precondition</i>	The manager is operating within a shared corporate business workspace.
<i>Result</i>	A new task record assignment appears on the target worker's workspace agenda
<i>Main Scenario</i>	1. Manager accesses the team workspace management view and selects "New Assignment".

	2. System queries and displays active team members linked to the workspace ID. 3. Manager assigns a specific worker row, attaches description briefs, and selects a deadline. 4. The system saves the task mapping variables inside PostgreSQL.
<i>Alternative Scenarios</i>	[At Step 3] The selected team member is no longer available in the workspace. 3a. The system validates that the selected team member still belongs to the workspace before saving the assignment. 3b. If the team member has been removed or deactivated, the system displays an error message and cancels the assignment.
<i>Non-Functional Constraints</i>	Performance: Delivering assignment notices to member dashboards should happen via WebSocket updates within 1 second.

4.2.19. Use Case 19

<i>Use case ID</i>	U019
<i>Use Case</i>	View Team Task Schedule
<i>Brief Description</i>	Renders a master team schedule overview calendar detailing all upcoming deadlines and assignments.
<i>Actor</i>	Manager
<i>Precondition</i>	The Manager is authenticated and has access to the target workspace.
<i>Result</i>	The calendar displays all scheduled tasks and upcoming deadlines for the current workspace.
<i>Main Scenario</i>	1. The Manager navigates to the Calendar module. 2. The system retrieves all scheduled tasks and deadlines for the current workspace. 3. The system displays a monthly calendar with scheduled tasks marked on their corresponding dates. 4. The system displays a Timeline panel containing upcoming tasks and deadlines.
<i>Alternative Scenarios</i>	[At Step 2] The system cannot retrieve the workspace schedule. 2a. The system displays an error message. 2b. The Manager may retry loading the calendar.

<i>Non-Functional Constraints</i>	Usability: The calendar interface shall provide a clear monthly calendar and timeline view, enabling Managers to quickly identify scheduled tasks and upcoming deadlines.
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4.2.20. Use Case 20

<i>Use case ID</i>	U020
<i>Use Case</i>	Post Review Management
<i>Brief Description</i>	A post approval process where the Manager reviews and evaluates content quality before it is published
<i>Actor</i>	Manager
<i>Precondition</i>	A post with the status "Pending Review" exists in the current workspace.
<i>Result</i>	The post status is updated to either "Ready for Distribution" or "Rejected", and the author is notified of the review outcome.
<i>Main Scenario</i>	1. The Manager opens the Post Management module. 2. The system displays all posts with the status. 3. The Manager selects a post and reviews its content. 4. The Manager approves the post. 5. The system updates the post status to Ready for Distribution. 6. The system initiates the publishing workflow. 7. The system notifies the post author that the post has been approved.
<i>Alternative Scenarios</i>	[At Step 4] The Manager rejects the post. 4a. The Manager enters feedback explaining the required revisions. 4b. The system updates the post status to Rejected. 4c. The system notifies the author and provides the review comments.
<i>Non-Functional Constraints</i>	Auditability: The system shall permanently record each review action, including the reviewer, timestamp, review decision, and comments for audit purposes.

5 Prototype

This section presents the visual design plan for **Omni Platforms**. It includes comprehensive, high-fidelity prototypes that establish the system's exact layout, color palettes, typographic hierarchy, and visual components. The goal of this phase is to translate core functional requirements into a tangible, production-ready interface design, allowing stakeholders to fully review and validate the user experience before software development commences. Specifically, this phase illustrates the design architecture of the application's core feature set, including the **Dashboard, Content, Statistics, Distribution, Calendar, Team Workspace, Prompt & Context, and Post Management** modules.

To build these polished user interfaces and interactive system maps, **Figma** was utilized as the primary design tool to craft every layout, graphical element, and responsive dashboard view. Architecturally, the platform bypasses a traditional marketing landing page in favor of an immediate, all-in-one web application dashboard. The global interface utilizes a standardized layout featuring a fixed left navigation sidebar coupled with a top utility bar containing a notification center and contextual account details (displaying the active user's name and assigned system role).

Technical Prototype Disclaimer

While the high-fidelity prototypes created in Figma present a complete visual design - incorporating the precise branding assets, user interface layouts, and color schemas intended for production - they remain a simulated environment to validate application mechanics and user flows. This blueprint is **not completely indicative of the final product's engineering implementation**, as minor visual adaptations, layout responsiveness variations, and front-end code optimizations may occur during the active development and integration phase.

a. Left Navigation Sidebar & Top Utility Bar



The application utilizes a standardized view architecture composed of a fixed left navigation sidebar and a global top utility bar, both framing the primary workspace canvas where the core feature interfaces are rendered.

- **Fixed Left Sidebar:** Accommodates the platform's core brand identity - currently rendered as text and architected to seamlessly support a future graphical logo asset - alongside an explicit menu array routing to the 8 primary system modules.
- **Top Utility Bar:** Serves as a persistent control layer housing dedicated access nodes for the systemic Notification Service and User Account Configuration parameters. This component additionally features a contextual workspace role indicator that dynamically updates to reflect the user's currently active permission tier (e.g., Manager, Member, or Personal).

Note: To optimize screen real estate and structural clarity, the following UI illustrations omit both the persistent left sidebar and the global top utility bar. Furthermore, the provided wireframe details the dashboard primarily through the **Manager view**, as it encompasses the most comprehensive feature set and administrative parameters. This view serves as the structural design baseline; interface variations for the Member and Individual roles are derived by dynamically modifying or omitting components from this master layout.

b. Dashboard



The dashboard utilizes a standardized view architecture composed of a fixed left navigation sidebar and a global top utility bar, both framing a primary workspace canvas. While the layout maintains a strict, unified design language to preserve visual continuity and minimize front-end rendering overhead, the system dynamically alters component contexts, widgets, and permission levels based on the active user role:

- **Persistent Structural Blueprint (Shared Layout):** Regardless of the active role, the main canvas features a personalized Greeting Card flanked by a Net Interaction Gain metric card and a Highest-Engaging Platform breakdown card, sitting directly above a horizontal Interaction Timeline Graph. The right-hand sidebar persistently anchors the My Calendar widget (individualized task scheduling) and the Channel widget (external connected networks).
- **Manager View (Baseline Master Layout):** Serves as the complete administrative control hub. The upper sidebar displays the Approval Requests queue; the lower canvas houses the Assigned Tasks ledger (mapping task names, deadlines, priorities, attachments, and assignees) positioned directly above the Recently Approved Posts

table; and the central analytics tier integrates a strategic **Team Goal Card** to monitor collective productivity.

- **Member View (Collaborative Production Layout):** Tailored for restricted team execution. It introduces a specialized metadata header displaying Workspace ID and Managed By fields; completely removes administrative configurations from the sidebar navigation; replaces the approval panel with a My Submissions Queue (showing *Pending*, *Approved*, or *Denied* states); swaps team targets for a Tasks Done progress bar; and splits the lower canvas into an individual My Tasks matrix positioned above a shared team Recent Posts activity log.
- **Individual View (Personal Creator Layout):** Optimized strictly for independent, solo creators. The upper sidebar block is completely omitted (sliding the calendar and channel widgets upward); the lower canvas displays a simplified My Tasks engine that strips out team-centric tracking parameters like "Assignee"; the secondary ledger block is entirely omitted as approval workflows do not apply; and the analytics tier integrates a personalized, self-set Goal for This Month counter card.

c. Content

Select Designated PlatformAdd >

☒ Facebook
The Discreet Coven

☐ Facebook
The Deep Blue Abyssal

☐ LinkedIn
Statch

PostEnable AI ☒

Enter the title

Have an idea but don't know how to write it properly?
Try our AI content generation feature from your original idea!

Manually enter your prompt or choose save onesGenerate

Drag & drop media here
or click to browse files (JPEG, PNG, MP4 up to 50MB)

Submit >

Save as Draft

Apply GEO/SEO

Knowledge BaseAdd >

choose your context

☒ BA Anual AnalysisView

☐ BA Anual AnalysisView

☐ BA Anual AnalysisView

Search

Prompt TemplateAdd >

choose your desired prompt

BA Anual Analysis
As a senior BA, put this business under close inspect...

BA Anual Analysis
As a senior BA, put this business under close inspect...

BA Anual Analysis
As a senior BA, put this business under close inspect...

Search

Select Designated PlatformAdd >

☒ Facebook
The Discreet Coven

☐ Facebook
The Deep Blue Abyssal

☐ LinkedIn
Statch

PostEnable AI ☐

Enter the title

Have an idea but don't know how to write it properly?
Try our AI content generation feature from your original idea!

Manually enter your prompt or choose save onesGenerate

Drag & drop media here
or click to browse files (JPEG, PNG, MP4 up to 50MB)

Submit >

Save as Draft

Apply GEO/SEO

Knowledge BaseAdd >

enable AI to use

☐ BA Anual AnalysisView

☐ BA Anual AnalysisView

☐ BA Anual AnalysisView

Search

Prompt TemplateAdd >

choose your desired prompt

BA Anual Analysis
As a senior BA, put this business under close inspect...

BA Anual Analysis
As a senior BA, put this business under close inspect...

BA Anual Analysis
As a senior BA, put this business under close inspect...

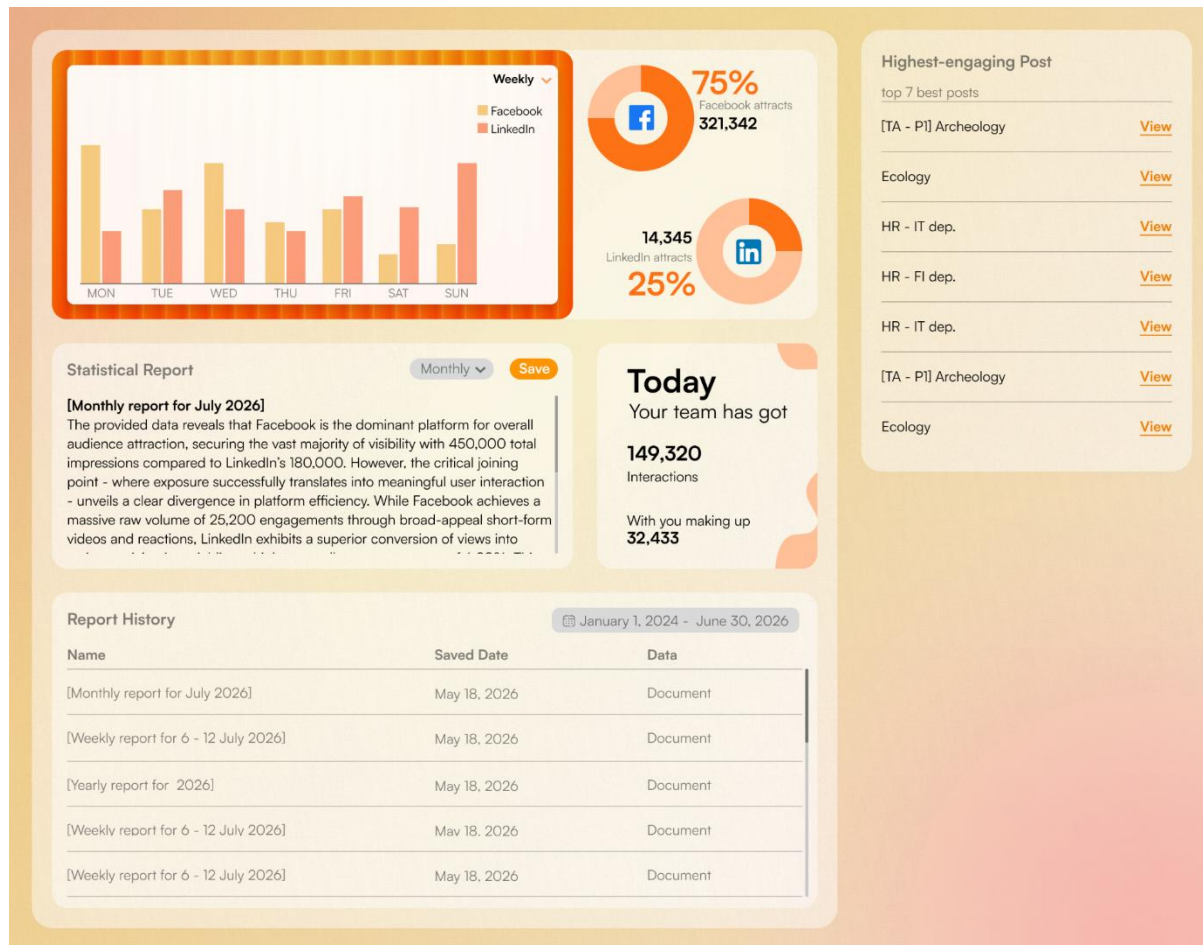
Search

The Create & Edit Post interface maintains a consistent layout structure across all three user roles (Manager, Member, and Individual). The workspace uses a persistent two-column layout framing a dynamic multi-tiered composition canvas on the left and an auxiliary AI context panel on the right.

The screen operates under two dynamic interactive states managed by a global **"Enable AI"** toggle switch in the upper-right corner of the editor card:

- **Left Content Canvas (Core Authoring Fields):**
 - Platform Selector: A horizontal array of card selectors to designate publishing targets (e.g., Facebook, LinkedIn).
 - Post Composition Fields: Houses fields for the Title, Content Body, and a Drag & Drop Media block supporting image/video uploads (up to 50MB).
 - Action Stack: Houses primary execution buttons - Submit, Save as Draft, and Apply GEO/SEO.
 - *AI State Behavior (Toggle ON):* Activates and displays a nested prompt composition bar and an orange Generate button directly inside the post editor canvas.
 - *Manual State Behavior (Toggle OFF):* Collapses and hides the prompt bar and Generate button entirely to maximize manual writing space.
- **Right Sidebar Panel (AI Resource Integration):**
 - Knowledge Base Panel: A searchable repository for uploading and storing organizational grounding documents (e.g., corporate analyses).
 - Prompt Template Panel: A searchable library of pre-configured system and custom generation prompts.
 - *AI State Behavior (Toggle ON):* Unlocks both panels, enabling interactive selection checkboxes to attach context files and templates to the AI generator.
 - *Manual State Behavior (Toggle OFF):* Grays out and disables both panels, rendering a helper message ("*enable AI to use*") to restrict interactive selection.

d. Statistics

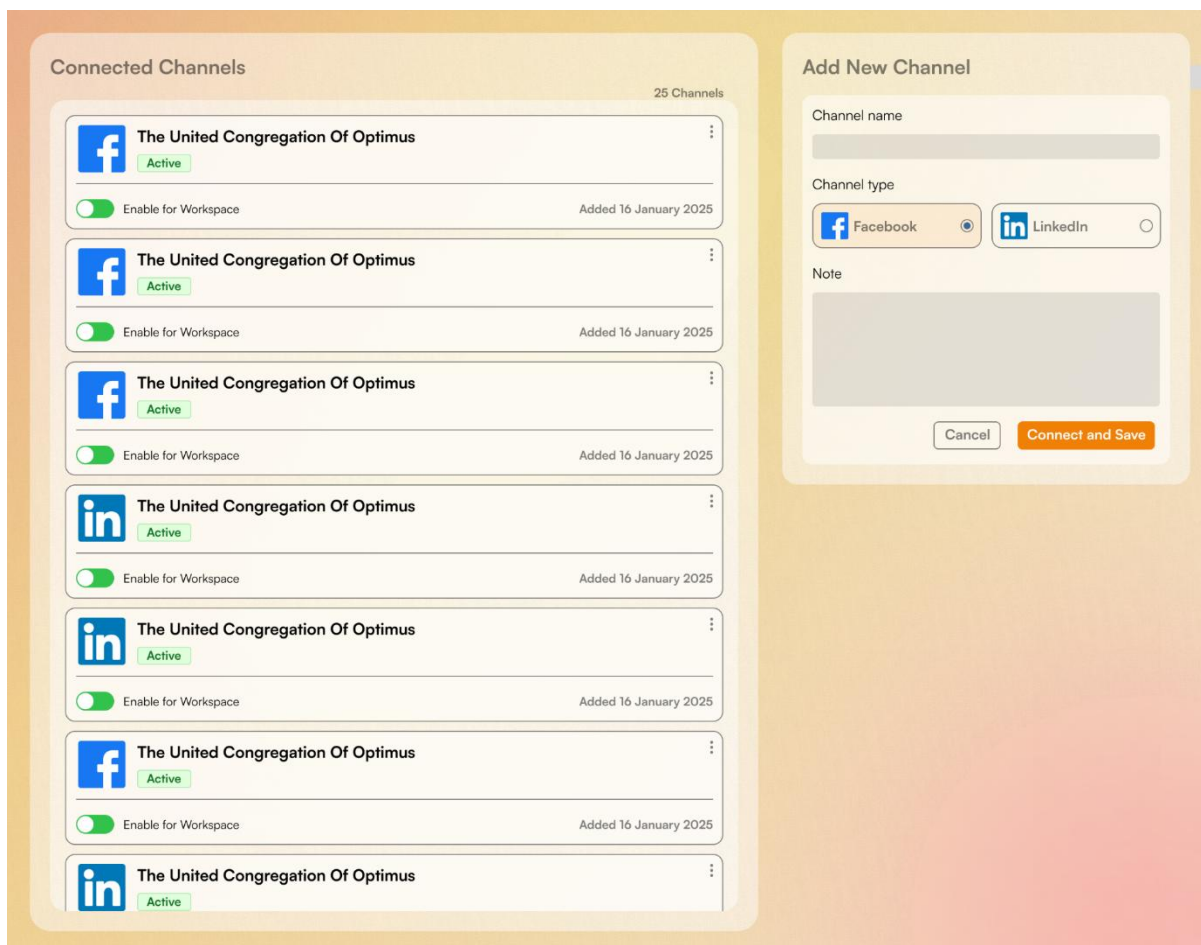


The **Statistical Report & Analytics** interface provides a comprehensive breakdown of multi-channel engagement metrics. The screen maintains a unified, consistent layout across all three user roles (Manager, Member, and Individual), featuring a multi-tiered data visualization canvas and a persistent sidebar. The only dynamic alteration across roles is the contextual scoping of the daily performance card:

- **Left Analytics Canvas (Core Visualization Panels):**
 - **Engagement Bar Chart:** A comparative, dual-axis bar chart displaying weekly performance trends across integrated platforms (e.g., Facebook and LinkedIn).
 - **Statistical Report Card:** A rich-text container displaying an AI-synthesized qualitative performance report, featuring a temporal filter dropdown (e.g., Monthly) and a **Save** action button.
 - **Report History Ledger:** A chronological data table visualizing historically archived reports with columns detailing the Report Name, Saved Date, and the corresponding Document reference.
- **Central Metric Tier (Distribution & Performance Indicators):**
 - **Platform Share Rings:** A set of dual donut-chart widgets representing proportional distribution shares (e.g., Facebook vs. LinkedIn) and total raw audience attraction numbers.

- **Dynamic "Today" Performance Card:** Features a core interaction counter and dynamically scopes its textual headers to reflect group versus individual contribution thresholds:
 - *Manager State:* Displays **"Today Your team has got"**, aggregating total workspace interactions alongside a sub-label detailing individual user contributions relative to the team total (e.g., *"With you making up..."*).
 - *Member State:* Dynamically transitions to **"Today You have got"**, filtering out all team-wide administrative data to strictly display the individual member's isolated performance metrics.
 - *Individual State:* Renders as **"Today You have got"**, focusing entirely on solo workspace metrics without any multi-user or collaborative baseline variables.
- **Right Sidebar Panel (Content Ledger):**
 - **Highest-Engaging Post Panel:** A persistent "Top 7 Best Posts" leaderboard, mapping high-performing content titles with direct **View** action buttons to inspect post-specific analytics.

e. Distribution

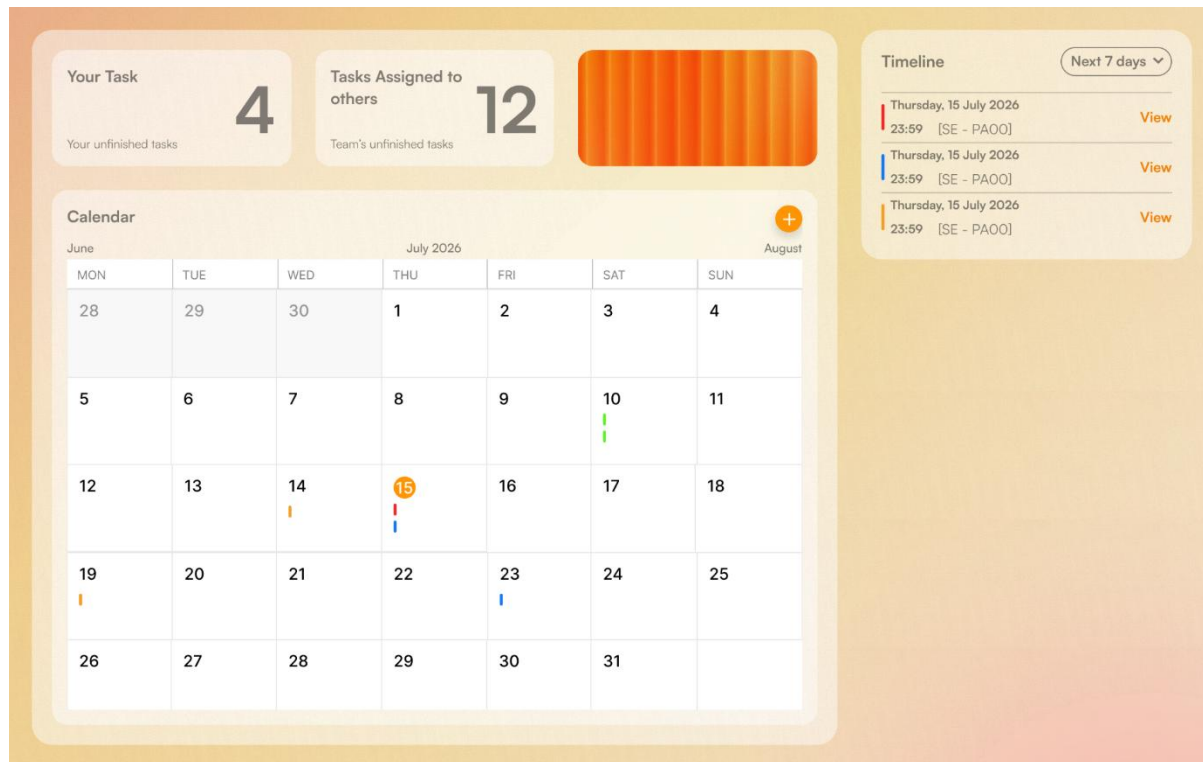


The **Channel Connection & Distribution** interface manages the integration of external social media platforms (e.g., Facebook and LinkedIn). The workspace is split into an expansive

integration list on the left and a dedicated connection sidebar on the right. Access to and the interactive capabilities of this screen are dynamically controlled based on the active user role:

- **Left Canvas (Connected Channels Ledger):**
 - **Core Channel Cards:** A scrollable vertical list detailing all currently linked social accounts. Each channel card displays the platform brand icon, verified channel name, a real-time status tag (e.g., *Active* or *Inactive*), the connection timestamp (e.g., *"Added 16 January 2025"*), and an options menu icon.
 - **Total Counter:** Displays the active network integration total (e.g., *"25 Channels"*) in the top-right corner of the container.
- **Right Sidebar Panel (Add New Channel Form):**
 - **Input Fields:** Houses structured fields to configure new accounts, including a text input for **Channel Name**, radio-card selectors for **Channel Type** (Facebook or LinkedIn), and a multi-line **Note** text area.
 - **Form Action Buttons:** Features persistent bottom actions-**Cancel** and **Connect and Save**-to process the OAuth pipeline and register the network.
- **Role-Based Access and Component Behavior:**
 - **Manager View (Complete Administrative State):** Displays the full baseline interface exactly as illustrated. Each card in the *Connected Channels Ledger* includes an active **Enable for Workspace** toggle switch, granting the Manager exclusive rights to authorize or restrict member access to specific publishing pipelines.
 - **Personal / Individual View (Solo State):** Renders both the *Connected Channels Ledger* and the *Add New Channel Form* for direct personal integrations. However, because this environment lacks collaborative workspaces, the **Enable for Workspace** toggle switch is completely omitted from all channel cards.
 - **Member View (Restricted State):** Denies access to this configuration interface entirely. Members do not possess the permission layer required to view, modify, or add channel credentials; they can only publish draft content to the pre-authorized channels enabled for their active workspace by the Manager.

f. Calendar



The **Task Calendar & Schedule Planning** interface provides users with an interactive, time-based overview of their deadlines and operational tasks. The workspace features an expansive calendar canvas on the left and a persistent, high-priority schedule sidebar on the right. While the general design framework is shared, the layout and data-tagging systems dynamically adapt to the active user role:

- **Left Calendar Canvas (Core Schedule Overview):**
 - **Interactive Monthly Calendar:** A comprehensive grid interface visualizing the current monthly cycle (e.g., July 2026) marked with color-coded task deadline indicators corresponding to specific dates. It includes a quick-action **Add (Orange "+")** button to schedule new deliverables directly.
 - **"Your Task" Metric Card:** A persistent, top-level KPI block showing the user's personal workload (e.g., "4"), with a sub-label reading *"Your unfinished tasks"*.
- **Right Sidebar Panel (Persistent Timeline Widget):**
 - **Timeline Widget:** A persistent vertical stream summarizing upcoming deadlines. It includes a configurable temporal filter dropdown (e.g., *"Next 7 days"*), chronologically ordered deadline entries displaying dates and times (e.g., *"Thursday, 15 July 2026 23:59"*), task ID tags, and direct **View** action links.
- **Role-Based Component and Data Behavior:**
 - **Manager View (Baseline Master Layout):** Serves as the complete collaborative control center exactly as illustrated. This view includes the **"Tasks Assigned to Others"** KPI card (e.g., "12"), displaying team-wide unfinished tasks. The timeline widget displays a full spectrum of colored tags: green (low priority), yellow

(medium priority), red (high priority), and **blue** (denoting active group or team tasks).

- **Member View (Collaborative Contributor State):** Adapts the workspace to center on individual responsibility within a team. The "**Tasks Assigned to Others**" KPI card is completely **omitted** to remove administrative clutter. The calendar grid and timeline display the full 4-color task coding system (including blue team tasks) to keep the member aligned with collective assignments.
- **Personal / Individual View (Solo Creator State):** Optimized strictly for independent workspaces. The "**Tasks Assigned to Others**" KPI card is completely **omitted** since no delegation mechanics exist in this role. To support isolated project tracking, the visual color-coding system for the calendar and timeline is filtered down to 3 individual tags: **green** (low priority), **yellow** (medium priority), and **red** (high priority). The **blue** tag is completely omitted from this view as team-based task categorization is irrelevant.

g. Team Workspace

The screenshot displays the 'Manage Your Workspace' interface. It features a 'Join Request' button at the top right. The main content is divided into several sections:

- Approval Request:** A table with columns: Task title, Submitted by, Attachment, Platform, and Action (Approve/Deny). It lists five requests, all for 'Architecture.png' attachments, with 'Approve' and 'Deny' buttons.
- Workspace Distributor:** A section with a 'Details' link and a list of four entries, each showing a Facebook icon, 'The Discreet Coven', and a toggle switch (three are on, one is off).
- Workspace Details:** A central banner with a gradient background. It displays: 'Marketing Dept.', 'Managed by: Harry Potter', 'Workspace id: 46345346', and 'Created: 8 February 2024'.
- Tasks Assigned to Others:** A table with columns: Name, Date, Priority, Attachment, and Assignee. It lists five tasks, all for 'Architecture.png' attachments, with priorities ranging from 'Low' to 'High'.
- Member Roster:** A sidebar on the right titled 'Member' with a 'Manage' link. It shows '45 Members' and a list of members, including 'Draco Malfoy' and several 'Fenrir Greyback' entries, each with a 'Joined 14 June 2025' date.

The

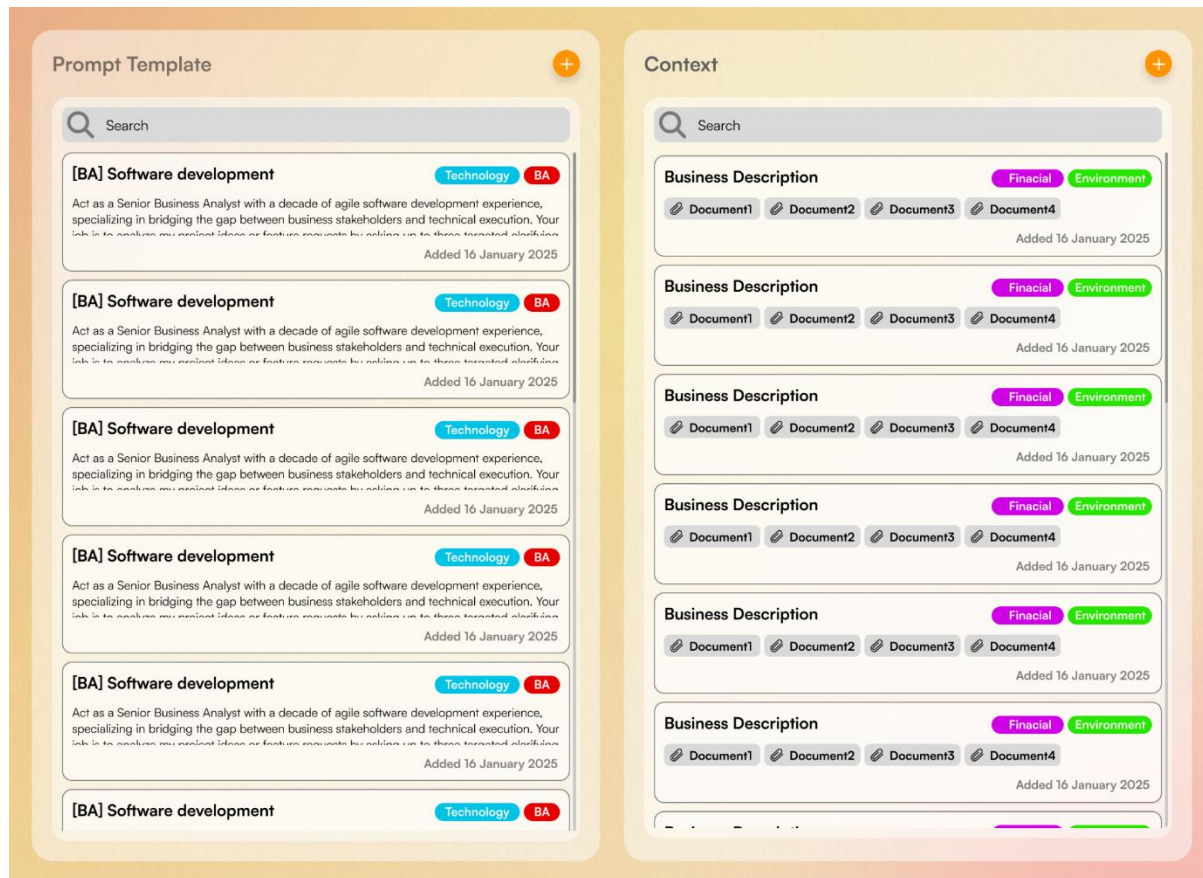
Workspace Management interface serves as the primary administrative dashboard for monitoring team collaborations, external publishing distributions, and task queues. The main view incorporates a left-aligned multi-widget workspace and a dedicated right-aligned team roster sidebar.

The access privileges and operational components of this view are dynamically modified based on the user's workspace role:

- **Persistent Structural Component (Shared Layout):**
 - **Workspace Details Card:** A centrally locked visual banner displaying essential metadata including the department title (e.g., *Marketing Dept.*), the active Administrator (e.g., *Managed by: Harry Potter*), the unique *Workspace ID*, and the creation timestamp. This component is identical for any role granted access to this interface.
- **Role-Based Access and Layout Variations:**

- **Manager View (Baseline Master Layout):** Grants full administrative authority exactly as illustrated:
 - **Header Actions:** Houses a clickable **Join Request** shortcut with an integrated red notification badge to handle pending member registrations.
 - **Approval Request Queue:** Lists submitted tasks detailing the Title, Author, Attachments, Target Platforms, and immediate administrative action nodes (**Approve / Deny**).
 - **Workspace Distributor Widget:** Features interactive **Enable/Disable** toggles next to each connected channel (e.g., Facebook, LinkedIn), giving the Manager direct authority to authorize or restrict channel usage inside the current workspace.
 - **Tasks Assigned to Others Ledger:** A broad management log highlighting delegated tasks, complete with deadlines, priority badges, and distinct assignee profile avatars.
 - **Member Sidebar Panel:** Lists all current workspace contributors (e.g., "45 Members"), accompanied by a fully interactive **Manage >** shortcut to update roster configurations.
- **Member View (Restricted Collaborative State):** Restricts the workspace view to focus on tracking individual output and verified statuses, stripping out executive overrides:
 - **Omitted Header Actions:** The **Join Request** administrative link and notification badge are completely hidden.
 - **Submitted Approval Requests:** The *Approval Request* queue transitions into **Submitted Approval Requests**, replacing the interactive *Approve / Deny* triggers with static, real-time workflow state labels.
 - **Workspace Distributor Read-Only Mode:** Toggles are completely removed. Instead, this widget displays static **Active** or **Inactive** status indicators (signaling whether the system API token remains valid). Members can only view the specific channels previously authorized by the Manager.
 - **My Tasks Ledger:** The *Tasks Assigned to Others* table is dynamically replaced with a personalized **My Tasks** matrix, filtering out other team members' workloads.
 - **Member Sidebar (View-Only Mode):** The list of active team members remains visible to preserve collaboration, but the administrative **Manage >** redirect shortcut is entirely **omitted**.
- **Personal / Individual View (Omitted Workspace state):** Because the concept of structured workspaces is strictly designed for enterprise environments, **this entire interface is completely omitted** and inaccessible for users operating under the solo Individual role.

h. Prompt & Context

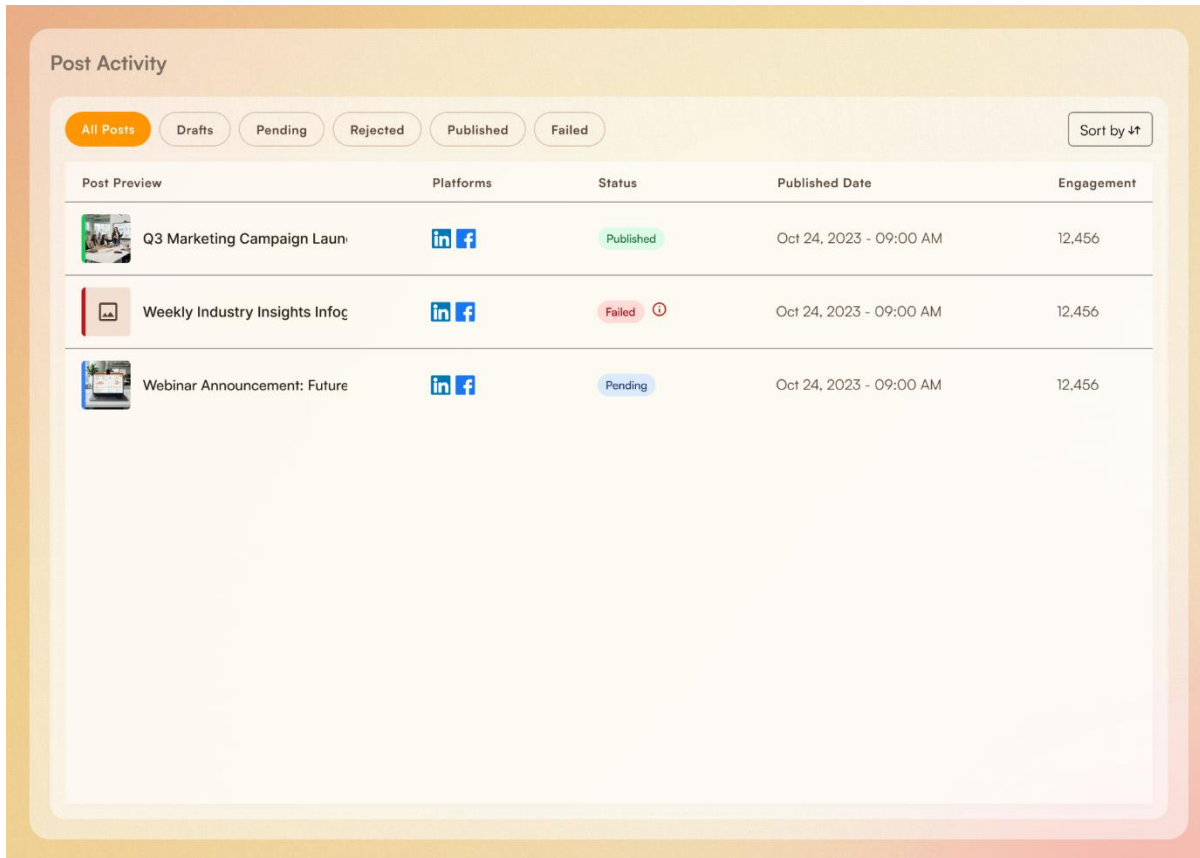


The **Prompt Template & Context Library** interface serves as a centralized asset management repository for organizing AI configuration tools. This specific screen maintains a completely uniform layout and structural blueprint across all three system roles (Manager, Member, and Individual), featuring a balanced two-column presentation canvas divided into independent, scrollable resource modules:

- **Left Component Canvas (Prompt Template Library):**
 - **Module Header:** Displays the **Prompt Template** section title alongside an interactive **Add (Orange "+")** shortcut button to initialize the creation of new templates.
 - **Search Integration:** Embeds a persistent contextual search field equipped with a magnifying glass icon to filter saved prompts by title, tag, or content string.
 - **Template Card Feed:** A scrollable vertical array of cards detailing individual generative parameters. Each template card houses a clear nomenclature title (e.g., [BA] Software development), color-coded classification tags (e.g., *Technology* and *BA*), a truncated plain-text preview of the underlying AI system prompt, and a historical creation timestamp (e.g., "Added 16 January 2025").
- **Right Component Canvas (Context Management Panel):**
 - **Module Header:** Displays the **Context** section title paired with a dedicated **Add (Orange "+")** action node to upload new baseline documentation structures.

- **Search Integration:** Houses a standard filtering input line to query saved contextual knowledge bundles.
- **Context Asset Feed(Knowledge Base):** A scrollable vertical stream of structured data cards mapping corporate data sets. Each container includes a core descriptor title (e.g., Business Description), thematic domain tags (e.g., *Financial* and *Environment*), a chronological registration timestamp, and an inline row of paperclip-indexed document attachment badges (e.g., *Document1*, *Document2*) verifying the uploaded text assets linked to that context profile.

i. Post Management



The screenshot displays the 'Post Activity' interface. At the top, there is a 'Post Activity' title and a filter ribbon with buttons: 'All Posts' (highlighted in orange), 'Drafts', 'Pending', 'Rejected', 'Published', and 'Failed'. A 'Sort by' button with an upward arrow icon is located on the right. Below the filter ribbon is a table with five columns: 'Post Preview', 'Platforms', 'Status', 'Published Date', and 'Engagement'.

Post Preview	Platforms	Status	Published Date	Engagement
 Q3 Marketing Campaign Launch	 	Published	Oct 24, 2023 - 09:00 AM	12,456
 Weekly Industry Insights Infoc	 	Failed 	Oct 24, 2023 - 09:00 AM	12,456
 Webinar Announcement: Future	 	Pending	Oct 24, 2023 - 09:00 AM	12,456

The **Post Activity** interface serves as a centralized historical ledger for tracking, auditing, and filtering content performance across multiple social networks. The interface maintains a uniform layout architecture across all three system roles, presenting an upper filtering panel above a structured data table. However, the available data filters and underlying status workflows dynamically adapt depending on the active user profile:

- **Persistent Structural Blueprint (Shared Design Layout):**
 - **Core Content Ledger:** A robust data table tracking historical publications. The ledger maps five mandatory criteria fields: **Post Preview** (featuring a small media thumbnail alongside the content title), **Platforms** (displaying distribution target icons, e.g., LinkedIn and Facebook), **Status** (color-coded workflow state indicators), **Published Date** (with precision timestamps), and **Engagement** (raw metric indicators).
 - **Utility Controller:** Embeds a persistent **Sort by** quick-action button in the top-right quadrant to reorganize the workspace rows chronologically or by engagement density.
- **Role-Based Filtering & Component Behavior:**
 - **Manager View (Baseline Master Layout):** Renders the complete dashboard matrix exactly as illustrated. The filter ribbon includes all six active options: *All Posts*, *Drafts*, *Pending*, *Rejected*, *Published*, and *Failed*. In this view, collaborative workspace status indicators utilize a distinct blue color-coding protocol to

highlight workflow checkpoints: **Pending** indicates team-submitted approval requests awaiting the Manager's review, while **Rejected** isolates items historically denied clearance by the Manager.

- **Member View (Collaborative Production State):** Mirrors the Manager's visual framework exactly as illustrated, featuring all six filter pill choices. While the UI mapping matches the master layout, the underlying database logic and textual context for the team-centric blue indicators switch orientation: **Pending** signals the member's own submitted drafts currently locked in the review queue awaiting supervisor action, while **Rejected** filters the subset of posts sent back to the member for revisions.
- **Personal / Individual View (Solo Creator State):** Optimizes the workflow for non-enterprise environments by completely removing team collaborative overhead. The filtering ribbon is dynamically reduced to a 4-pill state, displaying exclusively: **All Posts, Drafts, Published, and Failed**. The blue-coded team infrastructure tags (*Pending* and *Rejected*) are entirely **omitted** from this view, as solo creators bypass administrative review pipelines entirely.